

CSA0502 – QUERY PROCESSING FOR DATA SCIENCE

Lab Solutions – Set 4

Question 7

Write a Pandas program to create a Pivot table and find the maximum and minimum sale value of the items.

Python Code:

```
import pandas as pd

# Create realistic sales dataset
data = {
    'Region': ['East', 'Central', 'Central', 'West', 'East', 'Central', 'West', 'East'],
    'Manager': ['Martha', 'Hermann', 'Hermann', 'Timothy', 'Martha', 'Hermann', 'Timothy',
'Martha'],
    'Salesman': ['Alexander', 'Shelli', 'Sigal', 'Stephen', 'Diana', 'Luis', 'John',
'Karen'],
    'Item': ['Television', 'Home Theater', 'Television', 'Cell Phone', 'Home Theater',
'Television', 'Home Theater', 'Cell Phone'],
    'Units': [95, 50, 36, 27, 60, 75, 32, 60],
    'Unit_Price': [1198.0, 500.0, 1198.0, 225.0, 500.0, 1198.0, 500.0, 225.0],
    'Sale_Amount': [113810.0, 25000.0, 43128.0, 6075.0, 30000.0, 89850.0, 16000.0,
13500.0]
}

df = pd.DataFrame(data)

# Create Pivot table to find maximum and minimum sale value of items
pivot_max_min = pd.pivot_table(df, index='Item', values='Sale_Amount', aggfunc=['max',
'min'])

print("Pivot Table - Maximum and Minimum Sale Value per Item:")
print(pivot_max_min)
```

Output:

```
Pivot Table - Maximum and Minimum Sale Value per Item:
              max      min
Item
Cell Phone    13500.0    6075.0
Home Theater  30000.0   16000.0
Television   113810.0   43128.0
```

Question 8

Write a Pandas program to create a Pivot table and find the item-wise units sold.

Python Code:

```
import pandas as pd

# Create realistic sales dataset
data = {
    'Region': ['East', 'Central', 'Central', 'West', 'East', 'Central', 'West', 'East'],
    'Manager': ['Martha', 'Hermann', 'Hermann', 'Timothy', 'Martha', 'Hermann', 'Timothy',
'Martha'],
    'Salesman': ['Alexander', 'Shelli', 'Sigal', 'Stephen', 'Diana', 'Luis', 'John',
'Karen'],
    'Item': ['Television', 'Home Theater', 'Television', 'Cell Phone', 'Home Theater',
'Television', 'Home Theater', 'Cell Phone'],
    'Units': [95, 50, 36, 27, 60, 75, 32, 60],
    'Unit_Price': [1198.0, 500.0, 1198.0, 225.0, 500.0, 1198.0, 500.0, 225.0],
    'Sale_Amount': [113810.0, 25000.0, 43128.0, 6075.0, 30000.0, 89850.0, 16000.0,
13500.0]
}

df = pd.DataFrame(data)

# Create Pivot table to find item-wise units sold
pivot_units = pd.pivot_table(df, index='Item', values='Units', aggfunc='sum')

print("Pivot Table - Item-wise Units Sold:")
print(pivot_units)
```

Output:

```
Pivot Table - Item-wise Units Sold:
              Units
Item
Cell Phone      87
Home Theater   142
Television     206
```